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BIOMEDICAL LIMITED

ANNUAL PRODUCT QUALITY REVIEW

(YEAR – 2025)

Protocol No. : **BML/QA/APQR/QA/002**

Protocol Date : **23/04/2025**

Total No. of Pages :

PRODUCT NAME : Biomedical Paracetamol Syrup

GENERIC NAME : Paediatric Paracetamol Oral Solution

COMPOSITION : Each 5ml contains
Paracetamol BP 120mg
Flavour: Strawberry Flavour IN
Colour: Allura red

PACK SIZE : 60ml in an Amber Colour PET Bottle

PACKAGING TYPES AVAILABLE : 60ml Amber Colour PET Bottle duly labeled &packed in a Mono-carton.
Corrugated cardboard boxes carton

MASTER FORMULAR NUMBER : BML/QA/MFR/001SYP

SHELF LIFE: 3 Years

TIME PERIOD COVERED : JANUARY – DECEMBER, 2025

PRODUCT CODE : BL

BIOMEDICAL LIMITED

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APPROVAL PAGE

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1.0 SCOPE:

This covers internal annual products quality review, and all these parameters are monitored; changes in physical appearance of the product, chemical analysis (pH, Assay), microbiological analysis (Total Viable Count), and post market surveillance.

2.0 DESCRIPTION OF PRODUCT

Biomedical Paracetamol syrup is a red coloured liquid oral preparation in an Amber Color PET Bottle duly labeled and packed in a Mono-carton. Formulated for the treatment of pyrexia and mild to moderate; the clear solution's manufacturing process is carefully carried out to ensure the appropriate pH of the solution and a label claim of 120mg/5ml and consistency throughout the shelf-life is achieved. Careful control of all other parameters as mentioned in the scope and under the section for the reviews of the quality control of the product in this piece are assured at all times.

3.0 REVIEWS

3.1 REVIEW OF PRODUCT BATCHES MANUFACTURED

A batch of Biomedical Paracetamol Syrup 120mg/5ml comprises of about 41,666 bottles. Total No. of Batches produced within the period January 2025 to December 2025 is 30 batches i.e. from Batch no.:BL25001 to BL25030; all the batches produced have shown reliability of the manufacturing and the testing procedures in delivering constant and reproducible good product quality.

3.2 REVIEW OF STARTING AND PRIMARY PACKAGING MATERIALS USED IN THE FPP

The starting materials for the manufacture of Biomedical Paracetamol Syrup 120mg/5ml are Paracetamol powder BP (active ingredient), Citric Acid BP (excipient), Propylene Glycol (excipient), Ethanol (excipient), Sucrose (excipient), Sodium benzoate (excipient), SCMC (High -viscosity grade) (excipient), Allura red (excipient), Strawberry flavor (excipient) and Purified water BP (excipient). Paracetamol powder BP, Citric Acid BP, Propylene Glycol, Ethanol, Sucrose , Sodium benzoate, SCMC (High -viscosity grade) , Allura red and Strawberry flavor are sourced from approved vendors while Purified water BP is produced in house. The starting materials are each tested to comply with respective pharmacopeia and in-house specifications. The table below shows the different consignments of Paracetamol powder BP, Citric Acid BP, Propylene Glycol, Ethanol, Sucrose , Sodium benzoate, SCMC (High -viscosity grade) , Allura red and Strawberry flavor received and used in the manufacture of Biomedical Paracetamol Syrup 120mg/5ml within the year under review. All the batches of the active material received met the specifications of pharmacopeia.

3.2.1 VENDORS

All the raw materials used in the production of these batches are sourced from the approved vendors only.

3.2.2 RAW MATERIAL (API) MANUFACTURER/ SUPPLIER PERFORMANCE REVIEW

Name of API	Manufacturer (Name and Address)	Supplier (Name and Address)	Consignment / Batches Received	Consignment Approved	Consignment Rejected	Causes of Rejection
Paracetamol Powder BP	Liaoyuan City, China	Sagar Overseas, Nigeria	**3**	**3**	**Nil**	**Nil**
Citric Acid anhydrous BP	Shandong Ensign Industry	Sagar Overseas	**1**	**1**	**Nil**	**Nil**
Propylene Glycol	Shinghwa Amperex, China.	Bdv China.	**1**	**1**	**Nil**	**Nil**
	Dow Chemical Company	Sagar Overseas	**2**	**2**	**Nil**	**Nil**
	Tianjin Dongda Chemical	Sagar Overseas	**1**	**1**	**Nil**	**Nil**
	Gulshan Polyols Limited, Inidia	Sagar Overseas Limited, Lagos	**2**	**2**	**Nil**	**Nil**
	BUA foods (Sugar refinery), Lagos, State.	BUA foods (Sugar refinery), Lagos, State.	**1**	**1**	**Nil**	**Nil**
	Mebula Healthcare Limited	Sagar Overseas Limited.	**1**	**1**	**Nil**	**Nil**
	SK plc Global co Limted, South Korea.	Sagar Overseas Limited.	**1**	**1**	**Nil**	**Nil**
Sucrose	BUA Sugar, Refinery Limited, Lagos.	BUA Sugar, Refinery Limited, Lagos.	**7**	**7**	**Nil**	**Nil**
Ethanol	Greenfield Global.	Fiyique Ventures Ltd, Lagos	**3**	**3**	**Nil**	**Nil**
Sodium benzoate	Tianjin Dongda Chemical Group.	Sagar Overseas Limited, Lagos	**1**	**1**	**Nil**	**Nil**
SCMC(High viscosity grade)	Changzhou Guoyu Environmental Science, China	Bdv Behrens, Gmbh.	**1**	**1**	**Nil**	**Nil**
Allura red	Sun Food Tech, India.	Bdv Behrens, Gmbh.	**1**	**1**	**Nil**	**Nil**
Strawberry flavor	Lux Flavours.	Vitra Ventures	**1**	**1**	**Nil**	**Nil**

3.2.3 PRIMARY PACKAGING MATERIAL

The FPP is contained in 60 Amber Colour PET Bottle with 28mm ROPP cap bought as a finished product from the manufacturer of the PET Bottle. Manufacturer COA of the PET Bottle is reviewed and testing done in-house to confirm the quality specifications stated by the manufacturers as conform to in-house specifications set for the primary packaging material.

Name of Primary packaging Material	Manufacturer (Name and Address)	Supplier (Name and Address)	Consignments / Batches Received	Consignment Approved	Consignment Rejected	Causes of Rejection
60ml Amber Colour PET Bottle	Poly Products Nigeria Limited. Ogun State, Nigeria.	Poly Products Nigeria Limited. Ogun State, Nigeria.	22	22	Nil	Nil

3.3 REVIEW AND STATISTICAL ANALYSIS OF QUALITY CONTROL AND IN-PROCESS CONTROL

3.3.1 SPECIFICATIONS

Appearance: Deep Red Viscous Syrup
Paracetamol B.P.: 95.0% – 105.0%
pH: 3.80 – 6.80
Relative Density: 1.10g/ml – 1.20g/ml

3.3.2 ANALYSIS RESULTS:

The summary of analysis results of all the batches manufactured was prepared and the table showing the results presented. The summary of results for the chemical and microbial analysis of the batches manufactured is shown.

Chemical/Microbial Analysis for Finished product Result

S/N	BATCH NO	PARACETAMOL (95.0%–105.0%)	PH (SAMPLE) (3.80 – 6.80)	RELATIVE DENSITY (1.10-1.20g/ml)	pH Water (DM) (5.5 – 7.5)	BACTERIA (NMT10^3cfu/ml)	MOULD/YEAST (NMT10^2cfu/ml)	COLIFORM (Must be absent)
1	BL25001	99.65	4.69	1.12	6.43	00	00	COMPLIES
2	BL25002	100.7	4.79	1.12	6.43	00	00	COMPLIES
3	BL25003	100.37	4.79	1.11	6.42	00	00	COMPLIES
4	BL25004	100.12	4.81	1.11	6.45	00	00	COMPLIES
5	BL25005	100.24	4.84	1.11	6.47	00	00	COMPLIES
6	BL25006	100.9	4.82	1.11	6.42	00	00	COMPLIES

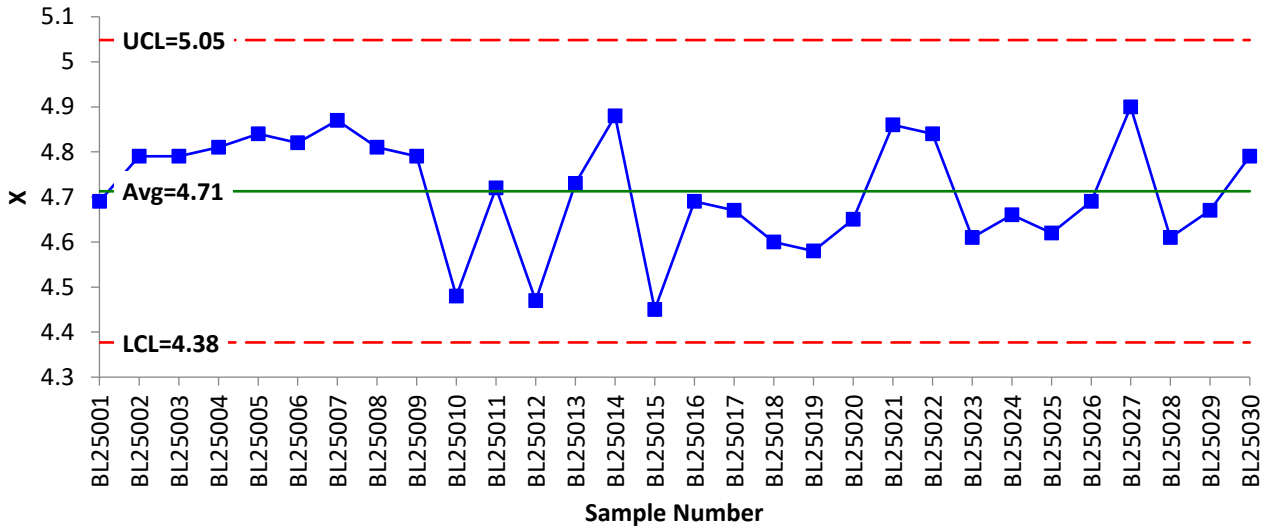
7	BL25007	99.66	4.87	1.1	6.41	00	00	COMPLIES
8	BL25008	100.46	4.81	1.11	6.43	01	00	COMPLIES
9	BL25009	99.53	4.79	1.1	6.42	00	00	COMPLIES
10	BL25010	100.36	4.48	1.11	6.4	01	00	COMPLIES
11	BL25011	99.4	4.72	1.1	6.44	00	00	COMPLIES
12	BL25012	100.23	4.47	1.12	6.23	00	00	COMPLIES
13	BL25013	99.76	4.73	1.1	6.45	00	00	COMPLIES
14	BL25014	100.22	4.88	1.11	6.4	00	00	COMPLIES
15	BL25015	100	4.45	1.11	6.4	04	00	COMPLIES
16	BL25016	98.24	4.69	1.11	6.4	01	00	COMPLIES
17	BL25017	100.23	4.67	1.11	6.45	00	00	COMPLIES
18	BL25018	99.88	4.6	1.11	6.21	00	00	COMPLIES
19	BL25019	99.64	4.58	1.1	6.51	00	00	COMPLIES
20	BL25020	99.65	4.65	1.1	6.19	00	00	COMPLIES
21	BL25021	100.58	4.86	1.11	6.31	01	00	COMPLIES
22	BL25022	99.88	4.84	1.11	6.44	00	00	COMPLIES
23	BL25023	98.34	4.61	1.12	6.52	00	00	COMPLIES
24	BL25024	99.19	4.66	1.12	6.63	01	00	COMPLIES
25	BL25025	99.66	4.62	1.1	6.4	02	00	COMPLIES
26	BL25026	99.76	4.69	1.1	6.39	02	00	COMPLIES
27	BL25027	98.12	4.9	1.1	6.11	00	00	COMPLIES
28	BL25028	98.28	4.61	1.11	6.13	00	00	COMPLIES
29	BL25029	99.21	4.67	1.12	6.74	00	00	COMPLIES
30	BL25030	101.25	4.79	1.12	6.81	00	00	COMPLIES

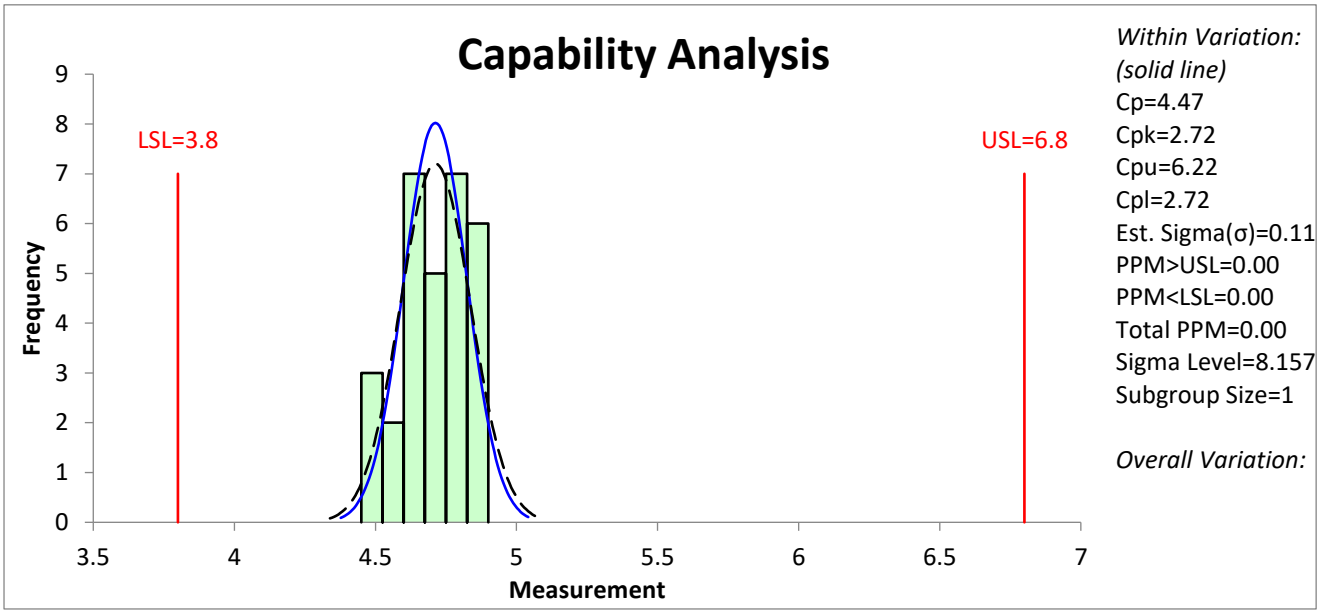
3.3.3 TREND ANALYSIS

Trend analysis of Water pH, Paracetamol content, Product pH, Total viable counts and Final yield %, were carried out on all the above batches.

Annexure I: pH trend for analysis of the product

Specification (3.8 -6.8**)**

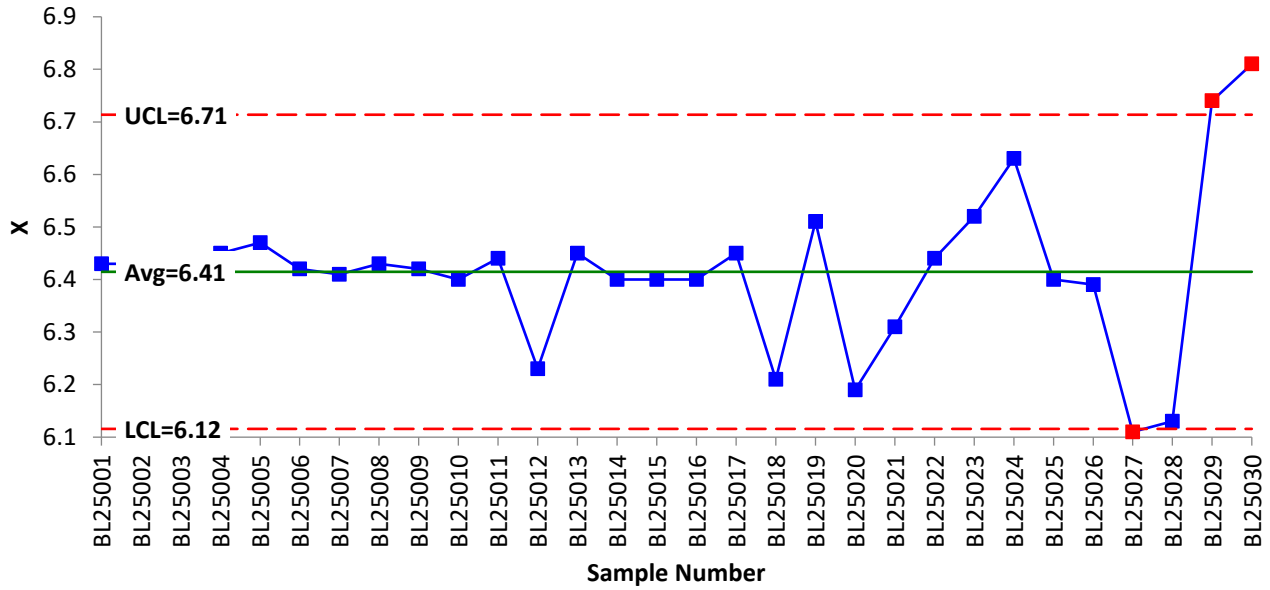


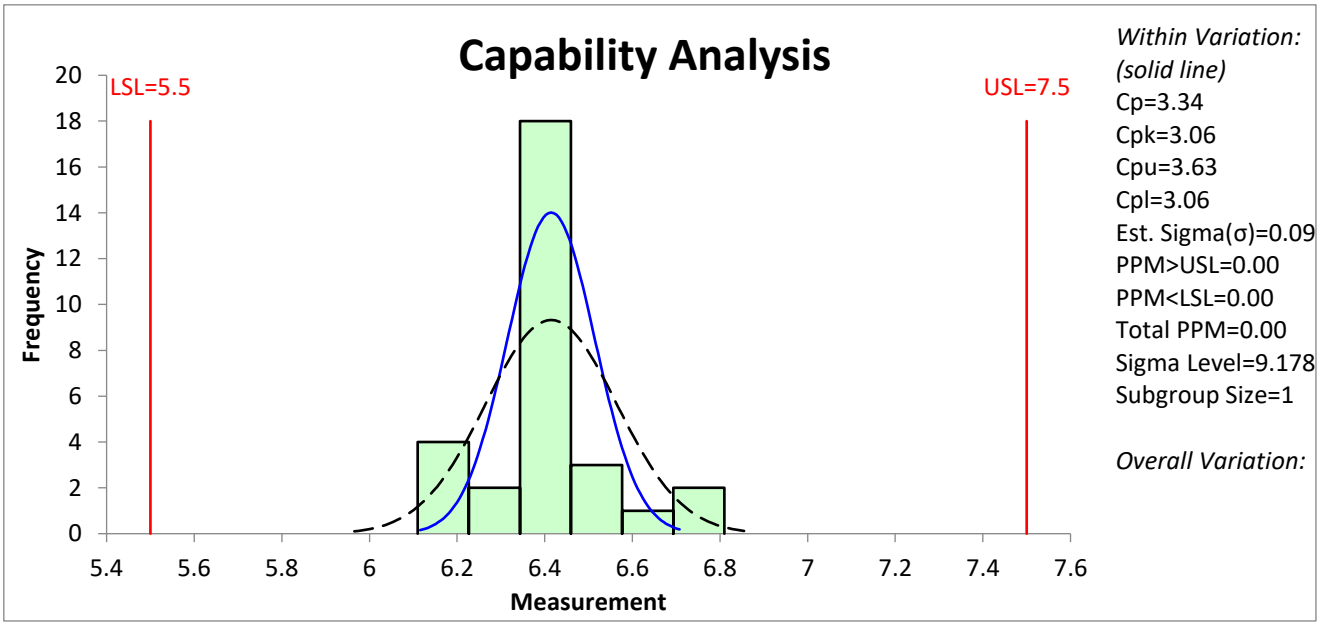


Mean	4.71
Standard deviation	0.124649
Maximum	4.9
Minimum	4.45
Control limit: 4.38 – 5.04	
Alert limit: –	
Minimum-Maximum: 4.45– 4.9	
Alert limit minimum well within the specification limit, control limit existing within the specification limit.	

Annexure II: pH trend of the Rinse Water (DM)

Specification (5.5 -7.5)

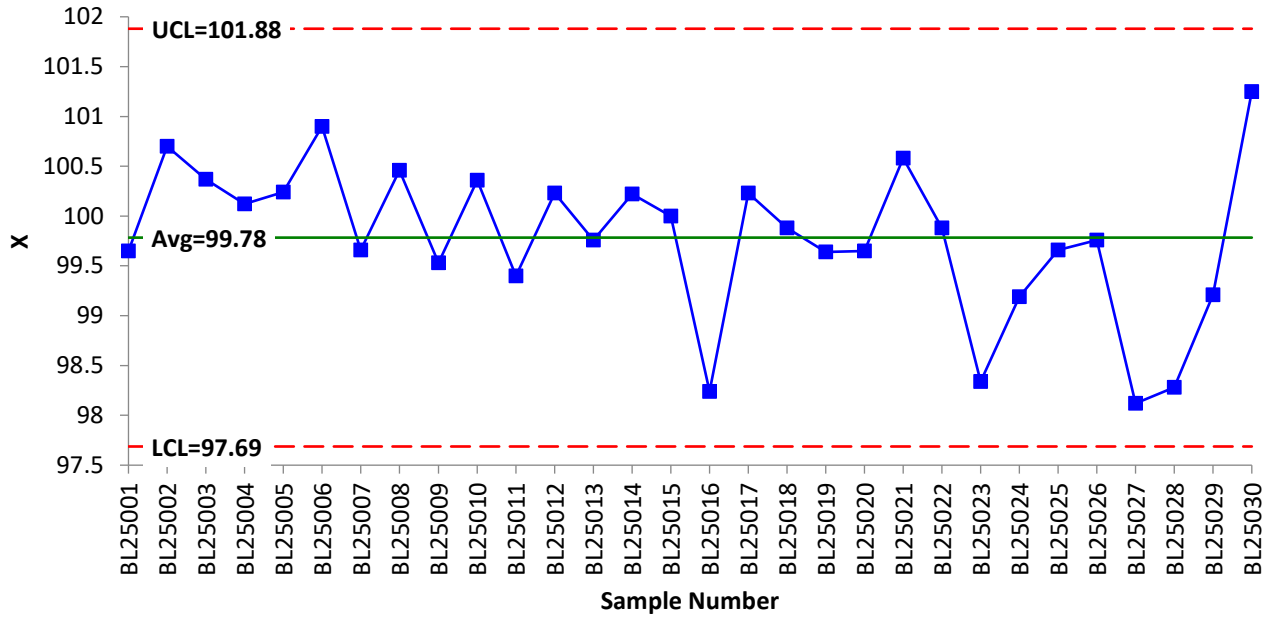


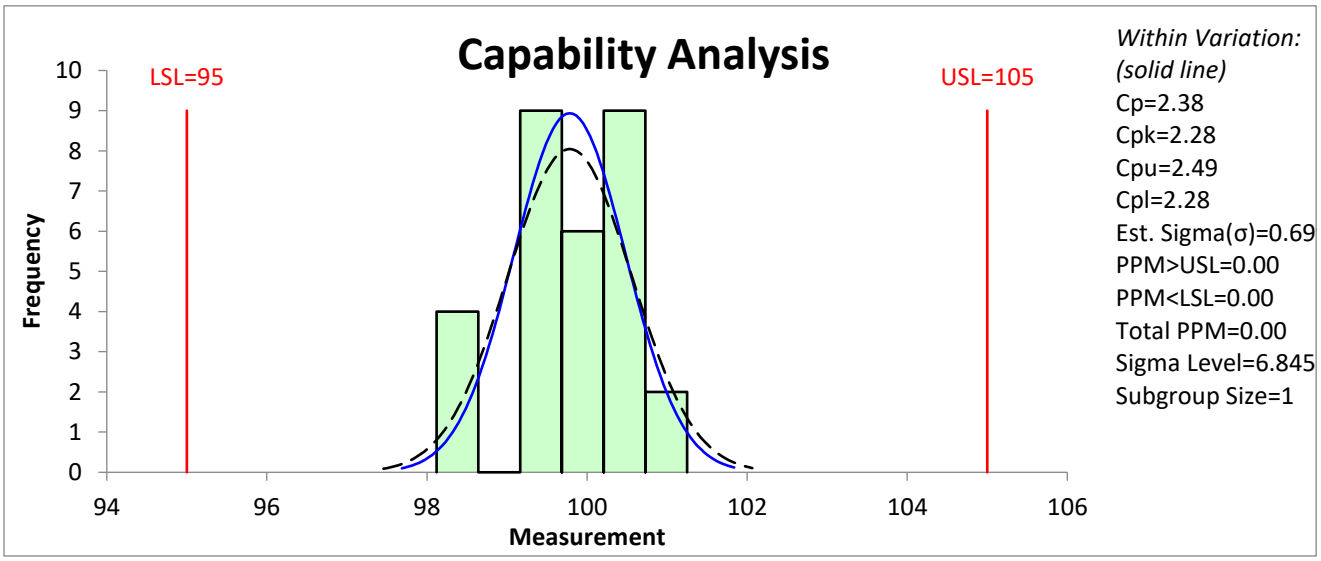


Mean	6.41
Standard deviation	0.149775
Maximum	6.81
Minimum	6.11
Control limit: 6.12 – 6.71	
Alert limit: –	
Minimum-Maximum: 6.11 –6.81	
Alert limit minimum well within the specification limit, control limit existing within the specification limit.	

Annexure III: Assay trend of analysis of Paracetamol in Biomedical paracetamol syrup.

Specification of Paracetamol (95 – 105%)

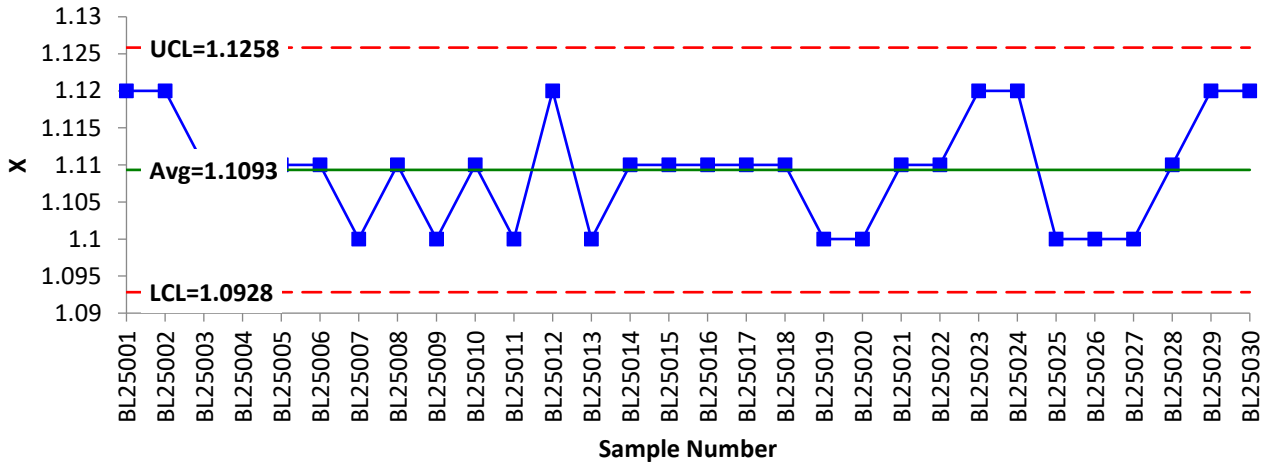


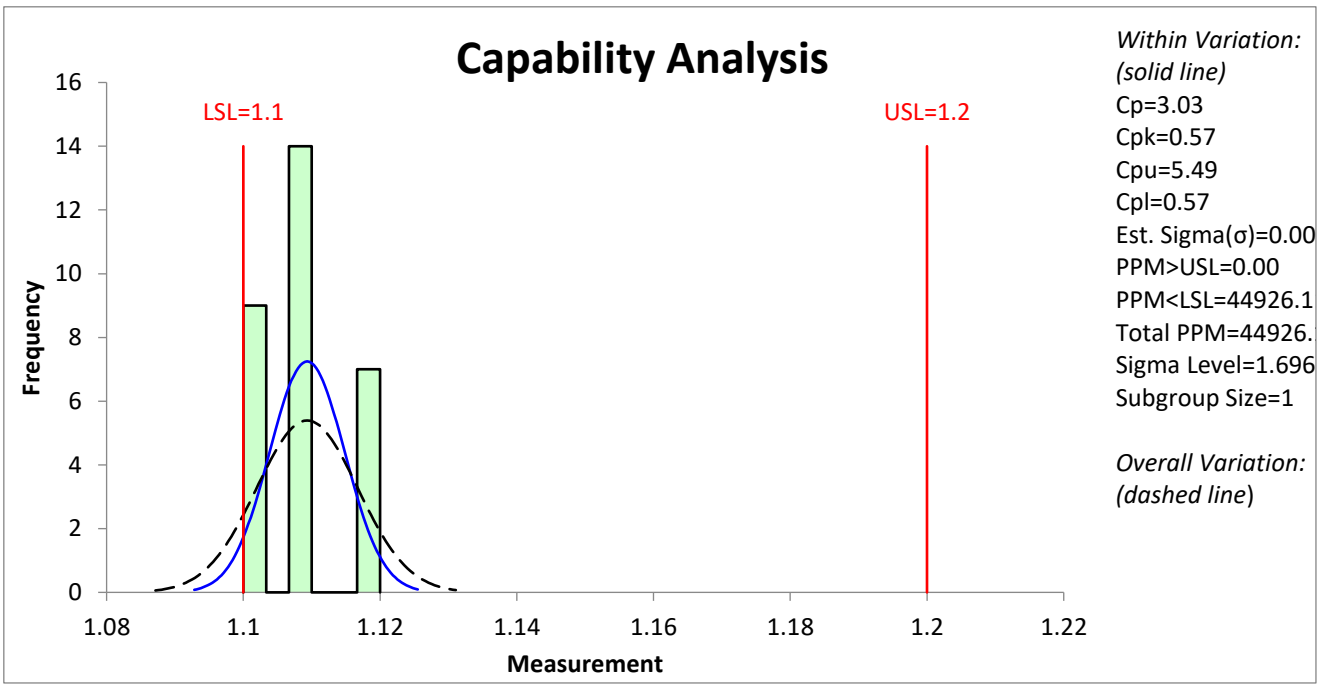


Mean	99.78
Standard deviation	0.775844294
Maximum	101.25
Minimum	98.12
Control limit: 97.69– 101.88	
Alert limit: –	
'Minimum-Maximum: 98.12 – 101.25	
Alert limit minimum within the specification limit, control limit existing along the specification limit.	

Annexure IV: Relative Density trend of product analysis of Biomedical paracetamol syrup.

Specification of Relative density (1.10-1.20g/ml**)**

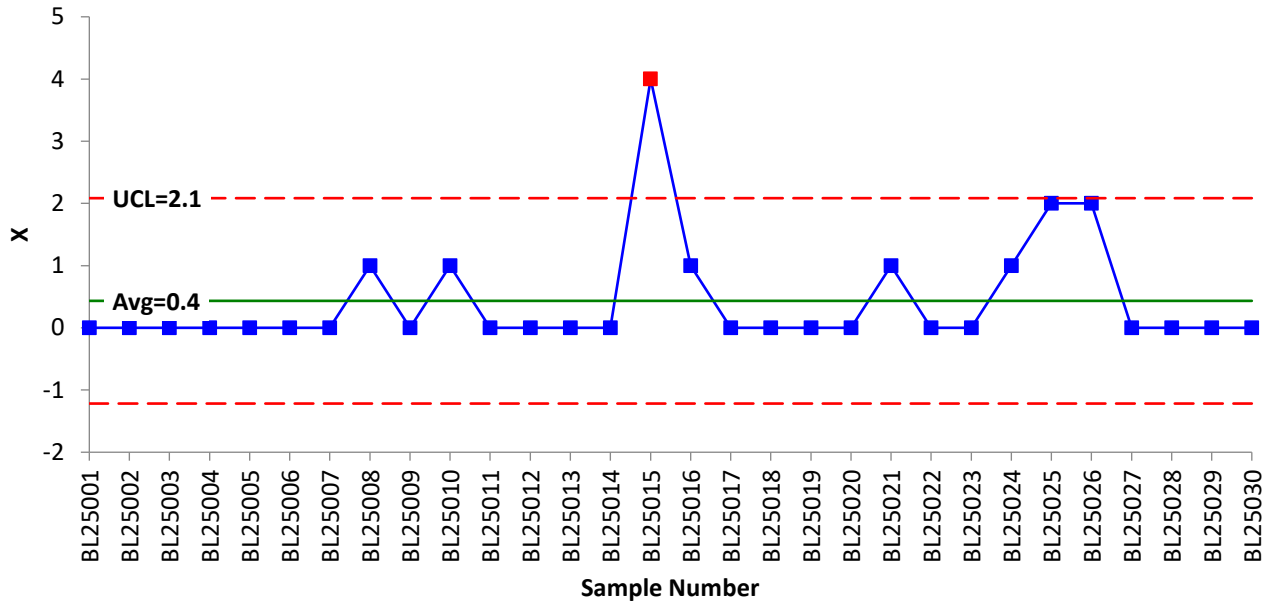




Mean	1.1093
Standard deviation	0.0073968
Maximum	1.12
Minimum	1.1
Control limit: 1.0928–1.1258	
Alert limit: –	
Minimum-Maximum: 1.1 – 1.12	
Alert limit minimum within the specification limit, control limit existing along the specification limit.	

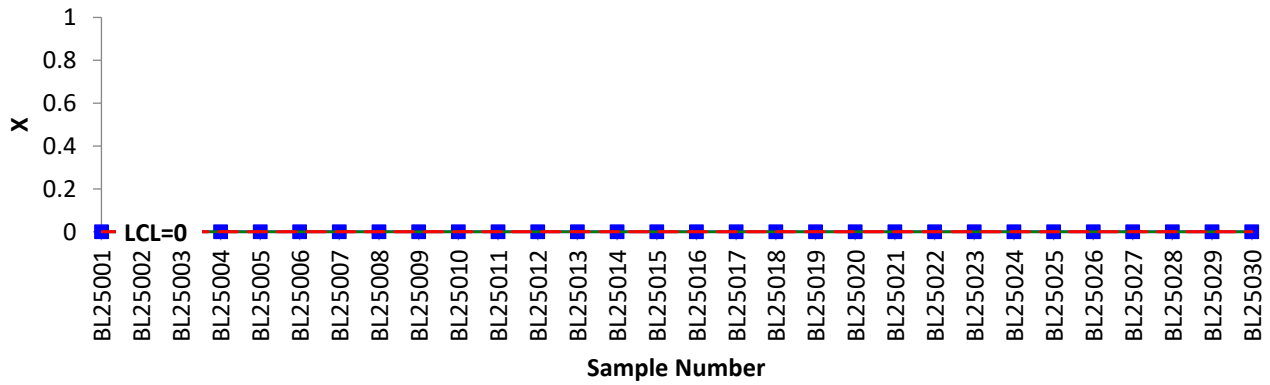
Annexure V: Trend of finished product analysis of Total Viable Count -bacteria in Biomedical paracetamol syrup.

Specification of Total Viable Count-bacteria (NMT 10^3Cfu/ml**)**



Annexure VI: Trend of finished product analysis of Total Viable Count –Mould/Yeast in Biomedical paracetamol syrup.

Specification of Total Viable Count-Mould/Yeast (NMT 10^2Cfu/ml**)**



SUMMARY OF PROCESS CAPABILITY

PROCESS CAPABILITY FOR PRODUCT			
1	pH		2.72
2	Relative Density		0.57
3	Paracetamol content		2.28
4	Rinse Water		3.06
5	Total Viable Counts	Bacteria	-
		Mould/Yeast	-

3.4 REVIEW OF BATCHES WITH OUT-OF-SPECIFICATION

No batch was recorded having out of specifications on any quality parameters.

3.5 REVIEW OF CRITICAL DEVIATIONS, NON-CONFORMANCES AND RELATED INVESTIGATIONS

3.5.1 PROCESS DEVIATIONS

No process deviation was reported from the standard processes for batches manufactured during the period Jan., 2025 to Dec., 2025.

3.5.2 YIELD DEVIATIONS

No observed deviation from the predetermined tolerance was observed.

3.5.3 NON CONFORMANCE

No nonconformance was reported from the standard processes for batches manufactured during the period Jan., 2025 to Dec., 2025.

3.5.4 BATCH FAILURES

No batch failure was reported from the standard processes for batches manufactured during the period Jan., 2025 to Dec., 2025.

3.6 CHANGE CONTROL

There was no change done to the processes, equipment or analytical methods for the manufacture of Biomedical Paracetamol Syrup during the year under review. As such, no change control was recorded throughout the period Jan., 2025 to Dec., 2025.

3.6.1 REGULATORY CHANGE

No regulatory changes as well as Master Formula Changes were made concerning these products throughout the manufactured batches till date.

3.6.2 CHANGES IN FORMULATION & PROCESS

No change is made in the formulation as well as process in regular practice. It has been observed that every batch manufactured, following consistent formulation and process, is giving consistent results which are proven by the trend analysis for all the parameters covered and expressed in graphical form

3.7 STABILITY MONITORING PROGRAMME

The results of the real-time and accelerated stability monitoring program of all the commercial batches carried out showed consistency of all Critical Quality Attributes (CQAs). The product's Critical Quality Attributes (CQAs) like pH, Assay, Relative Density, Appearance, Total Viable Count are all found to be within the set specifications in conformity with the limits of the pharmacopeia. The stability report is attached for reference as BML/QA/SBP/001 (2025)

3.8 QUALITY-RELATED RETURNS, COMPLAINTS AND RECALLS

3.8.1 MARKET COMPLAINTS

Up till now there is no market complaint received for this product during the period Jan. 2024 to Dec. 2024

3.8.2 RETURNED GOODS FROM MARKET

No finished goods were returned from the market for the period Jan., 2025 to Dec., 2025 due to any quality problem

3.8.3 BATCH RECALLS

Up till now there is no batch recall for product during the period Jan., 2025 to Dec., 2025.

3.8.4 SALVAGED GOODS

There is no salvaging of goods for the period of Jan., 2025 to Dec., 2025.

3.9 CORRECTIVE ACTIONS

No critical, major or minor deviations have been observed during the year under review. As such, no noteworthy corrective action was necessary. Few observed deviations during the year under review fall under the category of Others, which has to do with colour or print on tertiary packaging.

3.10 VALIDATED ANALYTICAL AND MANUFACTURING PROCEDURES

MANUFACTURING PROCEDURE

Equipment cleaning and sterilization
Dispensing of the raw material
Syrup preparation
Bottle Washing
Batch coding of Sticker label and mono-carton (packaging material)
Syrup filling
Final packaging of the Syrup

ANALYTICAL PROCEDURE

In-process testing of the cleanliness of equipment
In-process chemical testing of the semi-finished products for critical quality attributes like pH, assay and Relative Density.
Chemical testing of the finished products
Microbiological tests on the finished product

SUMMARY FOR PROCESS VALIDATION:

We have manufactured 30 batches of Biomedical Paracetamol Syrup 60ml with same manufacturing formula since Jan., 2025 till Dec., 2025 and yearly APQR data shows satisfactory trend throughout the period. Also the retained samples prove that the product manufactured with the existing formulation is stable

3.10.1.2 ENVIRONMENTAL CONDITIONS DURING MANUFACTURING OPERATION

The data of environmental conditions during the manufacturing of all batches were reviewed and recorded in the APQR. The environmental conditions of the critical areas were properly monitored in the raw material store, mixing unit, filling unit batch coding unit and storage areas

Environmental Conditions
Std Temp. : NMT 30°C
Relative humidity: NMT 70%

3.10.1.3 CRITICAL EQUIPMENT PERFORMANCE

The critical equipment used during the manufacturing and analysis of the product were identified and overall performance of these equipment were evaluated by reviewing the breakdown and preventative maintenance records.

3.11 REVIEW OF OBSERVED ADVERSE DRUG REACTIONS

No adverse drug reaction (ADR) was observed with the product during the year under review.

3.11 REVIEW OF PERIODIC RISK-BENEFIT EVALUATION REPORTS

Biomedical Paracetamol syrup is an antipyretic and analgesic oral solution formulated for the treatment of fever (pyrexia) and mild to moderate pain in pediatric patients, officially approved for sale in Nigeria since the year 2020. The periodic safety updates reports submitted for the products covering January-December, 2025 showed zero risk-benefit ratio (as there was zero risk recorded). All the columns under the spontaneous safety signal and risk report evaluation for the System Organ Class (SOC) are all showing zero. Over 1,000,000 bottles of Paracetamol syrup was manufactured and distributed across the nation in the year under review and no single report indicating either serious or non-serious early signal detection and/or safety risks on essential system organ class was recorded.

3.13 REVIEW OF POST MARKETING SURVEILLANCE

The reports of clinical and therapeutic performance of the products from major health institutions across the nation have proven the product to be safe and efficacious. No adverse reaction has been reported, no report of product contamination has been reported with the product within the year under review.

CONCLUSION

Process capability analysis was conducted on four physicochemical quality attributes of Paracetamol Syrup during the review period: pH, relative density, paracetamol content, and rinse water pH. Microbial attributes (TAMC and TYMC) were excluded from Cpk analysis, as count-based microbiological data are discrete and non-normally distributed, and do not satisfy the underlying statistical assumptions of capability indices; compliance for these parameters is assessed against pharmacopoeial acceptance criteria on a batch-by-batch basis.

Paracetamol content returned a Cpk of 2.28, indicating a process that is well-centred within specification limits and operating with substantial margin. This result demonstrates a high degree of manufacturing consistency for the primary active content across the review period.

Rinse water pH returned a Cpk of 3.06, reflecting a highly capable and consistent cleaning and rinsing process for the manufacturing tank. It is noteworthy that rinse water pH is not an isolated parameter; residual alkalinity or acidity carried over from inadequately rinsed tank surfaces can directly influence the pH of the next batch of Paracetamol Syrup introduced into that vessel. The strong capability observed for rinse water pH therefore has direct implications for product pH consistency, and sustained control of this parameter should be recognised as a contributing factor to overall product quality assurance.

Product pH recorded a Cpk of 2.72, which is statistically indicative of a capable and well-controlled process. The data suggest that pH across the review period remained consistently within the defined acceptance range with low variability. The acceptable rinse water pH performance noted above likely contributes to this outcome, and any future deterioration in rinse water control should be regarded as an early indicator of potential risk to product pH.

Relative density, however, returned a Cpk of 0.57, which falls significantly below the minimum capability threshold of 1.33 and indicates that the process is currently incapable of consistently meeting specification for this parameter. A Cpk at this level suggests either excessive process variability, a shift in process mean toward or beyond a specification limit, or both. This result is of concern from a product quality standpoint and requires formal investigation. A Corrective and Preventive Action (CAPA) has been initiated to identify root cause, which may include variability in excipient concentrations, inconsistencies in the dissolution or mixing step, inaccuracies in volumetric make-up, measurement system error at the point of density determination, or temperature effects on density readings.

Overall, three of the four physicochemical parameters demonstrated acceptable to excellent process capability during the review period. The relative density result represents the sole area requiring immediate quality system intervention, and progress against the associated CAPA will be monitored and reported in the next Annual Product Quality Review cycle.